

Elementary Problems

Angle Chasing

Version 1.0

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Basic Work

1. In a pentagon $ABCDE$, we have $\angle A = 135^\circ$, $\angle B = \angle C = 90^\circ$ and $\angle D = 72^\circ$. Find $\angle E$.
2. Find each interior angle of a regular hexagon.
3. In a convex hexagon $ABCDEF$, we have $AF \parallel CD$. Find $\angle A + \angle B + \angle C$.
4. In $\triangle ABC$, let D be the intersection of the internal angle bisector of $\angle CBA$ and the line through A parallel to BC . Prove that $AB = AD$.
5. In a regular pentagon $ABCDE$, a square $CDPQ$ is constructed inside the pentagon. Find $\angle BQC$.
6. Let M be the midpoint of the side BC of $\triangle ABC$. Prove that $\angle BAC = 90^\circ$ if and only if $MA = MB$.
7. (IMO 1959 Problem 5) An arbitrary point M is selected in the interior of the segment AB . The square $AMCD$ and $MBEF$ are constructed on the same side of AB , with the segments AM and MB as their respective bases. The circles circumscribed about these squares intersect at M and also at another point N . Let N' denote the point of intersection of the straight lines AF and BC .
 - (a) Prove that the points N and N' coincide.
8. Let O, A, B, C be collinear points in that order, and let O, X, Y, Z be collinear points in that order. If $AX \parallel BY$ and A, C, Z, X are concyclic, prove that B, C, Z, Y are concyclic.
9. In $\triangle ABC$, D is a point on the side BC . The circumcircle of $\triangle ABD$ meets the side AC again at E . The circumcircle of $\triangle ACD$ meets the side AB again at F . Let P be the intersection of BE and CF . Prove that B, D, P, F are concyclic, C, D, P, E are concyclic, and A, F, P, E are concyclic.
10. Given an acute $\triangle ABC$, we construct equilateral $\triangle XBA$, $\triangle YCB$ and $\triangle ZAC$ outside the triangle. Prove that the circumcircles of $\triangle XBA$, $\triangle YCB$ and $\triangle ZAC$ intersect at a single point.
11. Let $ABCD$ be a trapezoid with $AD \parallel BC$. The circumcircle of $\triangle ABC$ meet the side CD again at E . Prove that AB is tangent to the circumcircle of $\triangle DAE$.
12. In $\triangle ABC$, $AB < AC$. Let D be the foot of the internal angle bisector from A . The perpendicular bisector of AD meets the extension of CB at E . Prove that EA is tangent to the circumcircle of $\triangle ABC$.

13. Let $ABCD$ be a cyclic quadrilateral with $CB = CD$. The line through B parallel to CA intersects the line DA at E . Prove that $AB = AE$.
14. (Iranian Geometry Olympiad 2016 Medium) Let two circles C_1 and C_2 intersect in points A and B . The tangent to C_1 at A intersects C_2 in P and the line PB intersects C_1 for the second time in Q (suppose that Q is outside C_2). The tangent to C_2 from Q intersects C_1 and C_2 in C and D , respectively (the points A and D lie on different sides of the line PQ). Show that AD is bisector of the angle CAP .
15. (Iranian Geometry Olympiad 2016 Elementary) Let ω be the circumcircle of triangle ABC with $AC > AB$. Let X be a point on AC and Y be a point on the circle ω , such that $CX = CY = AB$. (The points A and Y lie on different sides of the line BC). The line XY intersects ω for the second time in point P . Show that $PB = PC$.
16. (Iranian Geometry Olympiad 2017 Medium) Two circles ω_1, ω_2 intersect at A, B . An arbitrary line through B meets ω_1, ω_2 at C, D respectively. The points E, F are chosen on ω_1, ω_2 respectively so that $CE = CB, BD = DF$. Suppose that BF meets ω_1 at P , and BE meets ω_2 at Q . Prove that A, P, Q are collinear.
17. (Greece MO 2017 Problem 1) An acute triangle ABC with $AB < AC < BC$ is inscribed in a circle $c(O, R)$. The circle $c_1(A, AC)$ intersects the circle c at point D and intersects CB at E . If the line AE intersects c at F and G lies in BC such that $EB = BG$, prove that F, E, D, G are concyclic.
(Remark: $c_1(A, AC)$ means the circle with centre A and radius AC .)

Finding Similar Triangles and Concyclic Points

18. In $\triangle ABC$, $\angle BAC = 90^\circ$. Let D be the foot of the altitude from A . Prove that $BD \times BC = BA^2$ and $BD \times CD = AD^2$.
19. Let E and F be points on the sides BC and CD of a square $ABCD$ respectively such that $BE = CF$. Prove that $AE \perp BF$.
20. In $\triangle ABC$, $\angle BAC = 90^\circ$ and $AB = AC$. Let D be a point on the side AB . A point E lies on the same side of the line CD as the point A . Suppose $\angle DEC = 90^\circ$ and $ED = EC$. Prove that $AE \parallel BC$.
21. Let P and Q be two points inside $\angle ABC$ such that $\angle ABP = \angle QBC$. Let X and Y be the projections of P on AB and BC respectively. Prove that $XY \perp BQ$.
22. (South Africa MO 2013 Problem 3) Let ABC be an acute-angled triangle and AD one of its altitudes (D on BC). The line through D parallel to AB is denoted by ℓ , and t is the tangent to the circumcircle of ABC at A . Finally, let E be the intersection of ℓ and t . Show that CE and t are perpendicular to each other.

23. Let $ABCDE$ be a convex pentagon such that $BCDE$ is a square with centre O and $\angle BAE = 90^\circ$. Prove that AO bisects $\angle BAE$.
24. Let A, B, C, D be collinear points in that order. A circle Γ with centre O passes through B and C . Prove that $OA = OD$ if and only if $AB = CD$.
25. In $\triangle ABC$, $AB = AC$. Let O be the circumcentre of $\triangle ABC$ and M the midpoint of BC . The perpendicular line from A to CO meets CO and the circumcircle of $\triangle ABC$ at D and E respectively. Prove that DM bisects EC .
26. (Bundeswettbewerb Mathematik 2016 Round 1 Problem 3) Let A, B, C and D be points on a circle in this order. The chords AC and BD intersect in point P . The perpendicular to AC through C and the perpendicular to BD through D intersect in point Q .
Prove that the lines AB and PQ are perpendicular.
27. (Flanders MO 2007 Problem 2) Suppose a semicircle with centre O and diameter AB is given. Let Z be a random point inside the semicircle, and let X be the intersection of OZ and the semicircle, and Y the intersection of AZ and the semicircle.
If P is the intersection of BY with the tangent line in X to the semicircle, show that $PZ \perp BX$.
28. (Bundeswettbewerb Mathematik 2007 Round 1 Problem 3) In triangle ABC points E and F lie on sides AC and BC such that segments AE and BF have equal length, and circles formed by A, C, F and by B, C, E , respectively, intersect at point C and another point D . Prove that the line CD bisects $\angle ACB$.
29. Given $\triangle ABC$ with $AB < AC$, points D and E lie on the sides AB and AC respectively such that $BD = CE$. The perpendicular bisector of BC meets the circumcircle of $\triangle ABC$ at M such that M lies on the same side of BC as A . Prove that A, D, E, M are concyclic.
30. (Portugal MO 2008 Problem 2) Let $AEBC$ be a cyclic quadrilateral. Let D be a point on the ray AE which is outside the circumscribed circle of $AEBC$. Suppose that $\angle CAB = \angle BAE$. Prove that $AB = BD$ if and only if $DE = AC$.
31. (Russian MO 1996 Grade 10 Problem 1) Points E and F are given on side BC of convex quadrilateral $ABCD$ (with E closer than F to B). It is known that $\angle BAE = \angle CDF$ and $\angle EAF = \angle FDE$. Prove that $\angle FAC = \angle EDB$.
32. (Iran MO 2nd Round 2007 Problem 1) In triangle ABC , $\angle A = 90^\circ$ and M is the midpoint of BC . Point D is chosen on segment AC such that $AM = AD$ and P is the second meet point of the circumcircles of $\triangle AMC$, $\triangle BDC$. Prove that the line CP bisects $\angle ACB$.

33. (Japan MO Finals 2006 Problem 1) Given five distinct points A, M, B, C, D in this order on the circumference of the circle O such that $MA = MB$, let P, Q be the intersection points of the lines AC and MD , and that of the lines BD and MC , respectively. If two intersection points of the line PQ and the circumference of the circle O are X, Y , then prove that $MX = MY$.
34. (IMO shortlist 2010 G1) Let ABC be an acute triangle with D, E, F the feet of the altitudes lying on BC, CA, AB respectively. One of the intersection points of the line EF and the circumcircle is P . The lines BP and DF meet at point Q . Prove that $AP = AQ$.
35. (Iranian Geometry Olympiad 2015 Advanced) Two circles ω_1 and ω_2 (with centres O_1 and O_2 respectively) intersect at A and B . The point X lies on ω_2 . Let point Y be a point on ω_1 such that $\angle XBY = 90^\circ$. Let X' be the second point of intersection of the line O_1X and ω_2 and K be the second point of intersection of $X'Y$ and ω_2 . Prove that X is the midpoint of arc AK .
36. Let $ABCD$ be a cyclic quadrilateral. A circle Γ passing through A and B meets the side CD at E and F so that E lies between C and F . The rays AF and BE meet at P . If $CB = CE$, prove that the line DP , the tangent at A to Γ and the perpendicular bisector of AP are concurrent.
37. (Russian MO 1995 Grade 9 Problem 2) A chord CD of a circle with centre O is perpendicular to a diameter AB . A chord AE bisects the radius OC . Show that the line DE bisects the chord BC .
38. (Bosnia Herzegovina TST 2013 Problem 1) Triangle ABC is right angled at C . Lines AM and BN are internal angle bisectors. AM and BN intersect altitude CH at points P and Q respectively. Prove that the line which passes through the midpoints of segments QN and PM is parallel to AB .
39. In $\triangle ABC$, let D and E be the midpoints of the sides AB and AC respectively. A point F lies on the circumcircle of $\triangle ADC$ such that A, D, C, F lie in that order on the circle. If $\angle BAF = \angle DEC$, prove that $AD \times FC = AF \times DC$.
40. Let AB be a diameter of a circle Γ . Consider any point X lying on Γ distinct from A and B . The tangent at X meets the tangents at A and B at Y and Z respectively. Prove that $AY \cdot BZ$ is a constant when X varies.
41. (IMO shortlist 1979) A circle $\mathcal{C}$ with centre O on base BC of an isosceles triangle ABC is tangent to the equal sides AB, AC . If point P on AB and point Q on AC are selected such that $PB \times CQ = \left(\frac{BC}{2}\right)^2$, prove that line segment PQ is tangent to circle $\mathcal{C}$, and prove the converse.
42. Let $ABCD$ be a convex quadrilateral with $\angle B = \angle D = 90^\circ$. Let P and Q be the projections of A and C on the diagonal BD . Prove that $BP = DQ$.

43. Let $ABCD$ be a cyclic quadrilateral. Its diagonals AC and BD intersect at E . Let M and N be the midpoints of AB and CD respectively. The extension of ME and EN meet the circumcircle of $\triangle CDE$ again at points P and Q respectively. Prove that $CD \parallel QP$.
44. (Turkey TST 1998 Problem 5) In a triangle ABC , the circle through C touching AB at A and the circle through B touching AC at A have different radii and meet again at D . Let E be the point on the ray AB such that $AB = BE$. The circle through A, D, E intersect the ray CA again at F . Prove that $AF = AC$.
45. In $\triangle ABC$, $AB = AC$. Let D be the midpoint of BC . Let E be the projection of D on the side AC . Let F be the midpoint of DE . Prove that $AF \perp BE$.
46. (IMO 2014 Problem 4) Points P and Q lie on side BC of acute-angled triangle ABC so that $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Points M and N lie on lines AP and AQ , respectively, such that P is the midpoint of AM , and Q is the midpoint of AN . Prove that lines BM and CN intersect on the circumcircle of triangle ABC .

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